Task-Scoped Composition of Agent Skills: Three Strategies Measured Against Loading the Whole Skill

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Abstract

An agent that needs three skills loads three whole documents, most of which do not apply to the task in front of it. We built a service that indexes a skill library once and returns, per request, a single composite scoped to one task and capped at a token budget, and we tested three ways of representing the library inside it. On a corpus we generated, with redundancy planted by construction, composition matched whole-skill loading within measurement error at 42% of the context ($\Delta = -0.022$ on a blind rubric, 95% CI $[-0.061, +0.018]$, $p = 0.30$). On 45 real skills in the Agent Skills format the same method lost 7.5 rubric points and 11 of 12 tasks ($p_{\text{Holm}} = 0.029$), and its cross-skill de-duplication was rejected by an independent adjudication panel on 56 of 60 sampled merges. Splitting each unit into a general *obligation* and the specific *lesson* that makes it actionable, and merging only the former while keeping every one of the latter, raised merge precision from 0.067 to 0.817 (difference +0.750, 95% CI $[+0.634, +0.866]$, $p < 0.001$) without improving the deliverables. An ablation shows why: printing the obligations costs quality, because a capable model already holds them. We retain the third representation and reject its rendering. Scaling the library from 45 to 504 skills, by adding 459 skills from twenty unrelated professional domains, leaves the composite unchanged in quality (+0.010, 95% CI $[-0.042, +0.074]$, $p = 0.95$) and in cost, while the alternative's cost grows linearly with the library because a skill library keeps every description resident. Total per-request context favours composition by 5.9 times at 45 skills and 25 times at 500. No variant matched whole-skill loading on output quality at any library size; the case for composition is that its cost does not move when the library does.

1 Introduction

Skill composition does not improve the work an agent produces. Across three strategies, 12 tasks and 156 blind-graded deliverables, no composite matched or exceeded loading the skills a practitioner would have chosen; the shortfall ranged from about 2 rubric points on a corpus built to contain the redundancy composition removes to about 12 points on a real library. What it does is hold its cost and its quality constant as the library grows, while the alternative's cost grows linearly: at 500 skills the total per-request context differs by a factor of 25. One of the three strategies also makes a library de-duplicable without losing content, and the number of obligations it can merge grows as $n^{1.53}$. Those results, and the reasons the quality gap persists, are what this article reports.

An agent skill is a document, typically a `SKILL.md`, that tells a model how a particular kind of work is done [Anthropic, 2025]. Skills are written to stand alone so they can be authored and shared independently. Two costs follow. A task drawing on three skills pays three times for whatever they share, and each skill is written for a class of situations, so most of what is

*Code, the demo corpus, the results and the scripts that regenerate every number in this article: `https://github.com/mathieubrglt/skillmerge`. The production corpus is not redistributed; see Section 9.

loaded does not apply to the one at hand; models attend less reliably to material buried in long context [Liu et al., 2024].

We built SkillMerge, a service exposed over the Model Context Protocol [Anthropic, 2024] that indexes a library once and returns a task-scoped composite. All model-dependent work happens at index time, so the request path is deterministic and involves no model call. Section 3 describes what is common to the three strategies and how they were evaluated. Sections 4 to 6 report the strategies in the order they were built, because each was a response to a measured failure of the one before. Section 8 states what we keep.

2 Related work

Retrieval-augmented generation [Lewis et al., 2020] established retrieving passages rather than whole documents. The move here is the same one applied to procedural rather than factual context, with two differences: the corpus is small, curated and written by hand, and the retrieved units must be assembled into something an agent can follow as instructions, which makes ordering and de-duplication part of the problem.

Prompt compression shrinks a prompt that already exists. LLMLingua [Jiang et al., 2023] and its successors [Pan et al., 2024] drop low-information tokens with a small language model, below the semantic level and on the request path. SkillMerge selects whole competencies above the semantic level and off the request path; a composite could then be compressed, so the two compose.

Our offline generation of predicted trigger phrasing is document expansion by query prediction [Nogueira et al., 2019] applied to procedural fragments, and it is what makes a purely lexical request path workable. Ranking uses BM25 [Robertson and Zaragoza, 2009]. Diversity reranking [Carbonell and Goldstein, 1998] appears in Strategy 1 and is removed in Section 4.5 because it did not earn its place.

3 Method

3.1 Two corpora

Synthetic (Strategy 1). Measuring de-duplication requires knowing what is duplicated, so we built a corpus from a ground-truth fragment library: 34 competencies in the earth-observation and commodity-intelligence domain, of which 10 were planted in five or six skills each, 12 in two to four, and 12 in one. Shared competencies were written in three paraphrases apiece, and 12 skills were rendered with per-skill headings, scaffolding prose and shuffled section order, so recovering the structure is an inference problem rather than an inversion of the generator. The result is 12 skills, 99 fragment instances over 34 distinct competencies, 10,137 tokens.

Real (Strategies 2 and 3). 45 skills exactly as their authors wrote them: 10 from an engineering plugin, 8 document-production skills shipped as platform defaults, and 27 example skills covering consumer tasks, design and tooling. They total 85,905 tokens, median 999 and maximum 9,921. Twelve ship auxiliary files, 200 of them. None was written with composition in mind.

3.2 Tasks, rubrics and grading

Each study uses a 12-task suite of realistic work requests, with the skills a practitioner would load recorded as an oracle set. The synthetic study's tasks sit in its own domain. The real study's suite spans two genres: seven engineering tasks (review a pull request, work a partial-failure incident, write an architecture decision record, judge release readiness, prioritise technical debt, chase a silent data-corruption bug, write a runbook) and five document-production

Study	Panels	Items	Weighted κ	Krippendorff α
1. Synthetic	2	468	0.913	0.904
2. Real library	2	1,200	0.940	0.923
3. Atoms	2	960	0.934	0.916

Table 1: Grader panel reliability. Exact agreement was 0.947, 0.955 and 0.951; every disagreement in all three studies was within one point.

tasks (tracked changes in a long contract, a deck from a report against a template, cleaning a malformed export, extracting line items from scanned invoices, planning an MCP server).

Every cell of every experiment was produced by an independent agent that saw only its prompt file, did not know which condition it was in or that other conditions existed, and received identical length guidance. Grading was blind: all candidates for a task were anonymised, shuffled and scored together on each rubric criterion at $\{0, 1, 2\}$, plus a 1 to 10 judgement of how useful the response would be to the person who has to act on it. Two independent panels graded every task.

Rubric provenance differs by study, and the difference matters. The synthetic study derived its rubric criteria from the same ground-truth fragments the composer indexes, which flatters any condition carrying those fragments. The real studies use two independently authored families: a *practitioner* rubric written by a panel that saw only the work request, and a *guidance* rubric written by a panel that saw the request plus the oracle skills. Both are scored and both are reported.

3.3 Statistics

Task-level scores are paired across conditions, so all contrasts are within-task differences. We report the mean difference, a bias-corrected percentile bootstrap 95% confidence interval over 20,000 resamples, the exact two-sided Wilcoxon signed-rank p computed over the full null distribution of sign assignments, Holm-corrected p within each study's family of contrasts, the paired standardised effect size d_z, the rank-biserial correlation, and the win/loss count over the 12 tasks. Twelve tasks is a small sample and we resist reading unadjusted p values.

Panel reliability is reported as exact agreement, quadratically weighted Cohen's κ and Krippendorff's α on the ordinal scale, all computed over the individual item scores. The merge-precision comparison in Section 6 is a two-proportion z test on independent samples of 60 pairs each.

Token counts use a byte-pair encoder; a second encoder agreed to within a fraction of a percent on sampled text. *Injected context* is the guidance block alone. *End to end* adds the task framing and the model's own output.

4 Strategy 1: fragment composition on a synthetic corpus

4.1 The method

Skills are split at second-level headings into fragments; per-skill scaffolding sentences are stripped by pattern. Fragments are embedded as L2-normalised TF-IDF vectors and clustered by average-linkage agglomeration on cosine similarity. Each cluster is a competency, represented by its shortest near-medoid member, which prefers the tersest phrasing of a shared obligation. For each cluster a model is asked once, offline, for eight short phrasings of situations in which it applies, plus six tags; the same is done for each skill description with fourteen phrasings. These are indexed at double weight. The expansion model sees only the fragment text and was never shown the evaluation tasks.

At request time, BM25 over skill descriptions and their expansions selects up to $k_{\max}$ skills scoring at least α times the top score. Each competency f is scored

$$s(f) \;=\; w_r\,\widehat{\mathrm{BM25}}(q,f) \;+\; w_o\,a(f) \;+\; w_s\,\frac{\log(1+\sigma_f)}{\log(1+\sigma_{\max})},$$

where $a(f)$ is the normalised routing score of the best skill containing f and σ_f is the number of skills it appears in. Competencies outside the routed set are kept at reduced weight, so a routing miss degrades rather than deletes. Selection is greedy under a hard budget on the rendered document.

4.2 Decomposition quality

The merging threshold was chosen on six skills and evaluated on the other six.

Split	k	true k	ARI	Precision	Recall
Development (6 skills)	28	27	0.967	1.000	0.938
Held out (6 skills)	27	25	0.904	1.000	0.829
Full corpus	35	34	0.956	0.985	0.930

Table 2: Cross-skill merging against ground truth at the development-selected threshold $\tau = 0.14$. Precision is over pairwise co-clustering decisions, so 1.000 means no two distinct competencies were ever merged.

Errors are one-sided: the method under-merges. That is the safer failure, since an unmerged duplicate wastes budget while a false merge silently deletes an obligation.

4.3 Coverage per token

Before asking whether agents write better deliverables, a cheaper question is available: for a given spend, how many of the competencies a task needs are present? On the eight tasks not used for hyperparameter selection, the composite saturates at 98.2% coverage for 1,272 tokens. Concatenating the oracle skills reaches 98.4% for 1,739 tokens, concatenating the routed skills 94.3% for 2,489, and loading the whole corpus 100% for 10,137. At the 1,000-token operating point the composite carries 0.87 covered competencies per thousand tokens against 0.57 for oracle concatenation and 0.38 for routed concatenation.

4.4 Deliverable quality

Three readings. The composite is not distinguishable from loading everything: the interval $[-0.061, +0.018]$ excludes any deficit larger than 6.1 rubric points at 42% of the injected context. This is non-inferiority within that margin and not superiority, since B wins 7 of the 12 tasks by small amounts. Second, selection rather than brevity is what works: the random control holds tokens constant and destroys only the mapping from task to content, and it loses 12.8 points while winning no tasks. Third, the saving is real end to end, since responses are also shorter (1,380 tokens against 1,467).

4.5 Ablations

At a 1,000-token budget on the held-out tasks, removing offline expansion costs 0.077 of coverage and removing cross-skill merging costs 0.049; removing both costs 0.182, more than the sum, because without merging the budget fills with duplicates and without expansion the wrong fragments are ranked into it. The other two components fail. Removing the diversity

Condition	Rubric	Useful	Ctx	E2E
A. No skill	0.711	7.00	0	1,370
D. Random fragments, token-matched	0.820	7.58	1,074	2,496
C. SkillMerge composite	**0.948**	8.46	1,098	2,623
B. Whole skills, concatenated	0.970	8.67	2,604	4,215

Table 3: Strategy 1, synthetic corpus. Rubric is the fraction of the maximum; *Useful* is the 1 to 10 usefulness judgement; *Ctx* is injected context tokens and *E2E* adds task framing and model output. $n = 12$ tasks, two panels.

Contrast	Δ	95% CI	d_z	p	p_{Holm}	W/L
C − A	+0.237	$[+0.169, +0.303]$	+1.92	0.0005	0.0024	12/0
C − D	+0.128	$[+0.069, +0.194]$	+1.12	0.0020	0.0059	10/0
C − B	−0.022	$[-0.061, +0.018]$	−0.31	0.3008	0.3008	2/7
B − A	+0.259	$[+0.195, +0.322]$	+2.23	0.0005	0.0024	12/0
D − A	+0.109	$[+0.061, +0.159]$	+1.18	0.0029	0.0059	10/1

Table 4: Strategy 1 contrasts on the combined rubric. Exact Wilcoxon signed-rank, Holm-corrected within the family of five.

rerank improves coverage by 0.025 and removing the support prior improves it by the same amount. Merging has already removed the near-duplication the rerank exists to suppress, so its penalty only displaces relevant material; the support prior is redundant with the routing term. Both are set to zero in later strategies.

4.6 What Strategy 1 does not establish

The corpus was built to contain the redundancy the method removes, and the rubric criteria were derived from the same fragments the composer indexes. Both choices make the result measurable and both flatter it. Strategy 2 repeats the experiment without either.

5 Strategy 2: the same method on a production library

5.1 Reading real skills

A real `SKILL.md` is prose interleaved with fenced code, tables of script invocations, ASCII diagrams, argument placeholders such as `@$1`, and links into `scripts/` and `references/` that mean nothing outside the skill's own directory. Fragments are therefore split into two classes at parse time. *Guidance* fragments are prose obligations and are mergeable. *Pinned* fragments are anything whose value depends on the skill it came from: a fenced block, a table, an invocation, or any text naming a file that ships with the skill. Pinned fragments are never merged, are offered only when their own skill is routed, and are carried verbatim with relative paths resolved to absolute ones.

Pinned material is 29.5% of fragments and 35.7% of corpus tokens. A composer that treats a skill as a bag of sentences, which Strategy 1 did because its corpus contained nothing else, destroys a third of a real library. From Strategy 2 onwards the composite is emitted as a valid `SKILL.md`, so nothing downstream needs a bespoke format and the output can be re-indexed by the reader that produced it.

Merging policy	Clusters	Cross-skill
None (raw fragments)	344	
Within-skill, raw text	200	0
Cross-skill, raw text	186	24
Cross-skill, normalised obligations	220	57

Table 5: Guidance fragments in the 45-skill library under each policy. Within-skill merging alone removes 42% of fragments.

5.2 There is less redundancy than expected

Within-skill merging collapses 344 guidance fragments to 200, a factor of 1.72 that is real and worth harvesting. Cross-skill merging on raw text adds 24 clusters across the entire library, and 6 inside the coherent ten-skill engineering sub-corpus. The synthetic corpus had a cross-skill redundancy factor of 2.91 because we put it there. Real skills mostly repeat themselves.

5.3 The normalisation trap

Raw-text similarity misses semantic overlap. Two skills can both require a rehearsed reversal path and share almost no vocabulary saying so. The available fix is to ask a model, offline and once, to restate each fragment as a single domain-free obligation, then cluster those.

Against ground truth this works. Calibrated on six skills of the synthetic corpus and evaluated on the other six, clustering normalised obligations reaches pairwise $F_1 = 0.953$ (ARI 0.952, precision 0.911, recall 1.000) against 0.907 for raw text, whose recall is 0.829. Every true co-clustering is recovered.

On the real library it fails silently. We sampled 120 cross-skill pairs, 60 merged by the obligation clustering and 60 near-misses just below threshold, and gave them to eight independent judges who were told nothing about how the pairs were chosen and asked whether a single well-written instruction could replace both without losing anything a practitioner would act on differently. Four of 60 merges were endorsed, none of the 42 high-confidence judgements among them, and mean judge confidence was 4.43 of 5.

Inspecting the rejects explains it. Normalisation was instructed to drop domain nouns, tool names and examples and keep the imperative. On a heterogeneous library that turns "use the design philosophy as the foundation for a single highly visual page" and "connector results arrive inline, so search wide and keep deliberately" into two generic sentences about being deliberate, which then cluster. The information that distinguished them is the information the normalisation removed. Fitting a conjunctive rule on obligation similarity and raw similarity together, selected on half the panel's labels and reported on the other half, reaches precision 0.111, because with four genuine duplicates in the sampled band no threshold can do better. Cross-skill merging is therefore off by default from Strategy 2 onwards, enabled only when a caller asserts a coherent single-domain corpus.

5.4 Routing is the ceiling

Skill routing over 45 skills reaches $F_1 = 0.674$ (precision 0.681, recall 0.667) against the oracle sets, close to the 0.662 the same router reached over 12. Three attempted improvements are worth recording because none worked. Aggregating each skill's best fragment scores into its routing score, a standard passage-evidence move, was neutral at every weight. Setting BM25's length normalisation b toward zero, on the argument that description length in this format tracks scope rather than verbosity, fixed individual cases without moving aggregate F_1. Hedging the routing width, narrow when the score distribution is peaked and wide when it

Condition	Pract.	Guid.	Both	Ctx	Recov.
A. No skill	0.910	0.694	0.802	0	
D. Random, token-matched	0.894	0.658	0.776	873	−23%
C. SkillMerge composite	0.871	0.808	0.840	967	33%
E. SkillMerge, token-matched to B	0.900	0.810	0.855	2,078	47%
B. Whole skills, oracle-selected	0.873	0.956	**0.915**	2,414	100%

Table 6: Strategy 2, real library. *Pract.* is the practitioner rubric, *Guid.* the guidance rubric. *Recov.* is the share of B's improvement over A that the condition recovers. Condition B is given oracle skill selection, which the composite is not, so the comparison is conservative.

Contrast	Δ	95% CI	d_z	p	p_{Holm}	W/L
C − B	−0.075	[−0.113, −0.033]	−1.00	0.0059	0.0293	1/11
E − B	−0.059	[−0.106, −0.014]	−0.70	0.0293	0.1172	1/9
E − C	+0.016	[−0.021, +0.051]	+0.24	0.4775	0.7607	6/5
C − A	+0.038	[−0.029, +0.109]	+0.29	0.3804	0.7607	8/4
C − D	+0.064	[−0.002, +0.132]	+0.51	0.0962	0.2886	9/3
B − A	+0.113	[+0.056, +0.177]	+1.00	0.0020	0.0117	10/1

Table 7: Strategy 2 contrasts, combined rubric. Only the two contrasts against whole skills survive Holm correction.

is flat, raised recall from 0.667 to 0.708 at a precision cost, and we kept it because fragment selection filters afterwards.

The residual failures are not tunable. One task describes a nightly job that "produced silently wrong totals ... runs fine, exits zero". The `debug` skill carries the offline expansion "output is wrong and i dont know why", which is semantically exact and lexically disjoint. No weighting of term statistics bridges that, and an embedding router is where the next gain lies.

What we could add is a way to fail quietly. The top skill's raw combined score, calibrated only on in-corpus probes (each skill's own expansions, short and concatenated-long) against out-of-corpus probes about cooking, football and gardening, gives a floor at which the server returns nothing. It keeps 95% of in-corpus queries and abstains on 14 of 15 out-of-corpus ones. All 12 evaluation tasks clear it, and the two lowest are the two the router gets wrong.

5.5 Deliverable quality

The Strategy 1 headline does not replicate. The composite gives up 7.5 rubric points and loses 11 of 12 tasks. Condition E raises the ceiling to the baseline's own token cost and lifts the relevance floor; it gains +0.016, indistinguishable from zero, and still loses to whole skills. The deficit is in what gets selected, not in how much room it has.

Where the two rubric families disagree is the informative part. On the practitioner rubric, written by a panel that saw only the request, *no skill* scores highest at 0.910: a capable model already writes a competent pull-request review. Skills earn their place on the guidance rubric, where whole skills reach 0.956 against 0.694 for no skill, and the composite reaches 0.808, about half the distance. A skill's value sits in its specifics, and specifics are what fragment selection drops first. Strategy 1, whose rubrics were all of the guidance kind, saw only the flattering half of this picture.

The genre split is usable. On document production, where skills carry script paths, library gotchas and format-specific procedure the model does not otherwise have, the composite recovers 51% of the benefit at 37% of the cost and beats both no guidance (0.833 against 0.733) and the random control (0.688). On engineering tasks it recovers nothing: 0.845 against 0.852 for no skill at all, because those skills largely restate practice the model already has and average 632 tokens, leaving nothing to save. Composition pays where a skill teaches and not where it reminds.

The random control is also informative. It scores below no guidance at all, recovering −23% of the benefit, so irrelevant procedural text is worse than silence.

6 Strategy 3: obligation and lesson atoms

6.1 The representation

Strategy 2's merging failure had a precise shape. To compare guidance across skills it normalised each fragment to a domain-free obligation, and to keep the budget it then kept one fragment per cluster and dropped the rest. The first step made unlike things look alike; the second made that permanent.

Strategy 3 stops conflating two things a fragment contains. An **obligation** is one imperative sentence: what the practitioner must do and when, in plain verbs with tool, product and file names stripped. It is the only field ever compared between skills. A **lesson** is the substance: the reason, the mechanism, the exact command, the failure it prevents, the names and numbers and paths, kept in the source's own words and verbatim where the source was verbatim. A lesson is never merged, never paraphrased, and never dropped when its obligation merges. A merged atom carries one obligation and every contributing lesson, each labelled with its source skill.

The question "can one instruction replace both?", which Strategy 2 had to answer yes to and could not, becomes "is this obligation a truthful heading for both lessons?", which is weaker and far more often true.

Skills cannot be used in this form as written, so Strategy 3 adds a reformatter: an offline pass in which a model reads one `SKILL.md` and emits its atoms, reproducing code blocks, command tables and paths exactly inside the lessons they belong to. Refactoring the 45 skills produced 1,134 atoms, 393 of them pinned, 20,311 tokens of obligation and 90,726 of lesson. The atomised library is 1.29 times the size of the source, an index-time cost paid once; obligations are 18% of it.

6.2 What refactoring costs

Measure	Value
Instructions preserved	98.8% of 588
Code blocks and command tables preserved	100%
File paths preserved	94.8%
Distortions (atoms state what the source does not)	49
Skills showing at least one distortion	9 of 11

Table 8: Reformatting fidelity, audited by independent agents reading the original and the atoms side by side.

Substance survives. Modality does not. Nearly every distortion was of one kind: the source's hedge became the atom's requirement. "Prefer using the imperative form" became "Write every instruction in the imperative form". A skill that says to ask when no mode is given acquired an atom demanding a mode. This is a predictable consequence of asking for imperatives, since obligations are grammatically mandatory and prose is not, so a rewrite into obligations is a rewrite into mandatoriness. A reformatter should be constrained to preserve must, should and may, and forbidden from introducing a requirement the source does not impose, and it should still be audited, because the pressure is structural.

One class of distortion is removable without a model. The reformatter emitted a bare `recalc.py` where the skill ships `scripts/recalc.py`. An invented path reads as authoritative and fails only when someone runs it, so every reference is now checked against files that exist, resolved when unambiguous and dropped otherwise: 135 kept, 110 dropped. Many of the drops are patterns rather than paths, such as `ppt/slides/slideN.xml`, so this is a conservative filter rather than an error count.

6.3 Merging, re-adjudicated

Under the atom design nothing is replaced by merging, so the corrected question asks whether the shared obligation is a truthful heading for both lessons, telling the judge explicitly that the two lessons need not resemble each other and that both are kept. An independent panel of eight judges saw 120 pairs on the same sampling design as Strategy 2.

	Strategy 2	Strategy 3
Merged pairs endorsed	4/60	**49/60**
Precision	0.067	**0.817**
Precision, judge confidence ≥ 4	0.000	0.808
Near-misses the panel would merge	0/60	5/60
Band-limited recall		0.907

Table 9: Cross-skill merge quality on the same 45-skill library, independent panels, mean judge confidence 4.5 of 5. Difference in precision +0.750, 95% CI $[+0.634, +0.866]$, $z = 8.3$, $p < 0.001$.

The merges are the ones a practitioner would want. *Obtain the person's explicit approval before the final irreversible action* is stated once and carries the lessons of eight consumer-task skills. *Assume the standard packages are installed and install one only after an import fails* carries the specific package, command and footgun list from `docx`, `pptx` and `xlsx`.

6.4 Deliverable quality, and the ablation that explains it

Solving the representation problem did not move the outcome. Condition F loses 4.1 rubric points to the fragment composite and a full point of rated usefulness on the 1 to 10 scale (-1.000, exact Wilcoxon $p = 0.008$ unadjusted), though on the combined rubric that difference does not survive correction.

If the atoms are right and the composite is worse, the suspect is what the composite prints, so we rendered the identical selection with the obligation headings removed and the freed budget spent on further lessons. Condition G recovers +0.021 on the rubric and +0.58 on rated usefulness (7.33 against 6.75), landing statistically level with the fragment composite. Printing the obligations was costing quality.

The explanation is Strategy 2's genre result turned on this design. Composition pays where a skill teaches and not where it reminds, because a capable model already holds the reminders. An obligation is by construction the reminder half: it is what remains once the specifics are stripped. The atom design spends 18% of the atomised library, and a comparable share of every composite's budget, restating in general terms things the model was going to do anyway, in the hardened modality of Table 8, which displaces specifics with commands.

We note the size of the effect honestly. +0.021 with an interval that includes zero is weak evidence on its own. It is supported by the direction being consistent across two measures, by the mechanism being independently motivated, and by G landing level with C while F does not.

Condition	Pract.	Guid.	Both	Ctx
B. Whole skills, oracle-selected	0.890	0.950	**0.920**	2,414
C. Strategy 2 fragment composite	0.867	0.808	0.838	967
F. Strategy 3, obligation and lesson	0.844	0.750	0.797	1,230
G. Strategy 3, lessons only	0.873	0.762	0.818	1,309

Table 10: Strategy 3. Condition F is worse than the fragment composite it was designed to replace, despite resting on merges twelve times more likely to be correct.

Contrast	Δ	95% CI	d_z	p	p_{Holm}	W/L
G − F	+0.021	$[-0.007, +0.050]$	+0.40	0.2500	0.6094	6/5
G − C	−0.020	$[-0.071, +0.031]$	−0.21	0.5049	0.6094	5/6
F − C	−0.041	$[-0.095, +0.007]$	−0.43	0.2031	0.6094	3/7
G − B	−0.102	$[-0.147, -0.060]$	−1.31	0.0024	0.0098	1/11
C − B	−0.082	$[-0.106, -0.058]$	−1.84	0.0005	0.0024	0/12

Table 11: Strategy 3 contrasts, combined rubric. After Holm correction only the contrasts against whole skills are distinguishable; the three composites are not separable from one another.

7 Scaling: what happens at 500 skills

Every result so far comes from a 45-skill library. Production libraries reach several hundred, and two things change there. A person can no longer read every description and pick, so the oracle selection that conditions B enjoyed is unavailable. And the descriptions themselves, which the Agent Skills format keeps resident so the model knows what it can load, become a per-request cost that is linear in library size.

We tested both.

7.1 A 504-skill library

The 45 real skills keep their real atoms and expansions. To them we added 459 synthetic skills across twenty unrelated professional domains, from hospital administration to aviation maintenance to library science, each with a description, eight trigger phrasings and four atoms, so that distractors compete for routing *and* for lesson selection rather than sitting inert. The twelve evaluation tasks are unchanged.

Routing does not collapse. Recall is flat at 0.667 from 100 skills upward and precision settles near 0.57, so F_1 falls from 0.673 to 0.614 between 45 and 200 skills and then stops falling. Offline expansion is what holds it up: the distractors have expansions too, and the router still finds the right skills. The router also narrows as the library grows, from 3.2 skills per request to 2.3, because more competition makes the score distribution peakier. Composition degrades gracefully rather than sharply: about a fifth of a composite's lessons come from distractors at 504 skills. Latency stays near a second on one core.

Merging moves the other way. Sub-sampling the real library at sizes from 5 to 45 skills, the number of obligations shared across skills grows as $n^{1.53}$ ($R^2 = 0.999$), and the lessons filed under them as $n^{1.60}$. Merge opportunity is pairwise, so a larger library gives any obligation more chances to recur: 3.0 merged obligations per skill at 45, and an extrapolated 11.2 at 500. The de-duplication argument gets stronger as the library grows, which is the opposite of how it behaved when we moved from the synthetic corpus to the real one.

Library	Units	Merged	Route F_1	Skills/req	Lesson purity	ms/req
45	872	137	0.673	3.2	1.000	12
100	1,081	134	0.668	2.8	0.928	15
200	1,445	164	0.608	2.5	0.835	882
300	1,809	198	0.610	2.4	0.808	1,253
400	2,156	239	0.614	2.3	0.822	1,120
504	2,524	284	0.614	2.3	0.790	1,170

Table 12: Routing and composition against library size. *Lesson purity* is the share of a composite's lessons drawn from the 45 real skills rather than from distractors. False abstentions were zero at every size.

Condition	Pract.	Guid.	Both	Ctx
Whole skills, oracle-selected (45)	0.885	0.952	**0.919**	2,414
Whole skills, router-selected (45)	0.865	0.875	0.870	5,594
SkillMerge composite (45 skills)	0.883	0.756	0.820	1,309
SkillMerge composite (504 skills)	0.867	0.794	0.830	1,193

Table 13: Blind rubric scores with the library at 45 and at 504 skills. Both whole-skill baselines use the 45-skill library, which favours them: their cost is measured before the resident-description term of Table 15.

7.2 Quality does not degrade

Growing the library by a factor of eleven does not move the composite: $+0.010$, interval $[-0.042, +0.074]$, $p = 0.95$. Its cost does not move either (1,193 tokens against 1,309). This is the property that matters, and it is a property of the design rather than of this corpus: the budget is a ceiling, so output size cannot grow with the library, and abstention keeps a query the library does not cover from producing anything at all.

Removing the oracle costs the baseline 4.9 rubric points ($p = 0.018$ unadjusted, 0.053 after correction) and more than doubles its tokens, from 2,414 to 5,594, because an uncertain router loads several whole skills where it would have loaded one. On the worst task it loaded 18,581 tokens. The composite's gap to the realistic baseline is 4.0 points and no longer distinguishable after correction; its gap to the unavailable oracle baseline stays at 8.9 points.

7.3 Where the argument turns

Composition is a bad deal on a small library and a good one on a large library. Quality stays flat in library size for the composite and stays slightly better for whole skills at every size we measured, so the crossover comes from the denominator. A skill library keeps its descriptions resident so the model can choose, and at 500 skills that is roughly 30,000 tokens on every request before anything useful is loaded. The composite replaces it with a tool description and a capped call.

So the answer to whether SkillMerge improves results at scale is no, and the answer to whether it improves at scale is yes. Output quality is invariant to library size. Cost is invariant to library size. The alternative's cost is not.

8 The retained system

We retain Strategy 3's representation and reject its rendering. The shipped configuration indexes the library as obligation and lesson atoms, merges across skills on obligations only,

Contrast	Δ	95% CI	d_z	p	p_{Holm}	W/L
504 − 45 (composite)	+0.010	[−0.042, +0.074]	+0.10	0.9492	0.9492	4/7
504 − routed whole	−0.040	[−0.099, +0.039]	−0.31	0.0742	0.1484	3/9
45 − routed whole	−0.050	[−0.076, −0.025]	−1.08	0.0049	0.0195	1/10
routed − oracle whole	−0.049	[−0.099, −0.013]	−0.61	0.0176	0.0527	1/9
504 − oracle whole	−0.089	[−0.124, −0.052]	−1.35	0.0020	0.0098	1/11

Table 14: Scaling contrasts, combined rubric, Holm-corrected within the family.

Library	Whole skills	SkillMerge	Ratio	Points/1k tok
10	6,194	1,406	4.4	0.14 vs 0.58
45	8,294	1,406	5.9	0.11 vs 0.58
100	11,594	1,406	8.2	0.08 vs 0.58
250	20,594	1,406	14.6	0.04 vs 0.58
500	35,594	1,406	25.3	0.02 vs 0.58
1,000	65,594	1,406	46.6	0.01 vs 0.58

Table 15: Total per-request context: resident skill descriptions (median 60 tokens each) plus what gets loaded. SkillMerge pays one 97-token tool description and one capped composite, so its column is constant. Points per thousand tokens uses the measured rubric scores of Table 13.

keeps every contributing lesson, and emits lessons alone, holding the obligation in the selection trace where a person debugging a composition can see it. Three guards from Strategy 2 stay, each of them the result of something that went wrong.

Abstention. Out-of-corpus queries previously produced confident irrelevance. The server now returns nothing when no skill scores above the calibrated floor, which catches 14 of 15 negatives while keeping 95% of positives.

Never worse than the source. A composite is capped at 90% of the cost of the skills it drew on, so composing cannot cost more than loading them. Without it, on small skills, the greedy selector filled a generous budget with unrelated material.

Pinned material carried verbatim. Code, tables, invocations and file pointers are never merged or paraphrased, and relative paths are resolved and validated against files that exist.

The composite is emitted as a valid `SKILL.md`, so no consumer needs a bespoke format and the output can be re-indexed by the reader that produced it. Composition on the request path stays deterministic and model-free: it is a pure function of the index, the query and the configuration, which makes it reproducible, auditable back to its source skills, and cheap enough to call before every task.

We recommend this configuration on two grounds, and not on a third. It reduces injected context by 46% to 60% at a quality cost that is real and bounded, and it makes a library deduplicable and searchable: one obligation, every skill that imposes it, every lesson attached. It does not improve deliverables, and we found no configuration that did.

9 Availability

The implementation, the twelve-skill synthetic corpus, every results file and the scripts that regenerate the tables in this article are at `https://github.com/mathieubrglt/skillmerge` under the MIT License. This article is available under CC BY 4.0. Running `tools/verify_article.py` checks all 325 reported numbers against the stored results and exits non-zero on any mismatch; `tools/stats_all.py` regenerates every contrast, effect size and reliability coefficient.

The 459 synthetic distractor skills of Section 7 are ours and are released with the rest. The 45-skill production library is not redistributed. Seventeen of those skills carry their own licence header and six are marked proprietary, and the atoms are close derivatives of them by design, since a lesson reproduces its source verbatim. The repository instead carries `docs/SKILLS-MANIFEST.json`, which records every skill the study read with the SHA-256 of the document as it stood, so a reader with the same installation can confirm they are looking at the same corpus before rebuilding the index locally. The synthetic corpus is regenerated byte-identically from its seeded generator; the calibration numbers reported for it in Sections 4 and 5.3 are stored alongside it in `results/` rather than recomputed by a script in this release, and the real-library half of Section 5.3 additionally depends on the withheld production corpus and its judge panel, so it cannot be rebuilt from what ships here.

10 Threats to validity

Sample size. Twelve tasks per study. Confidence intervals are over tasks, not over corpora or domains, and after Holm correction the differences among composites in Strategy 3 are not distinguishable. We rest the retained recommendation on the merge-precision result, which is a two-proportion comparison on 120 independently judged pairs, rather than on the composite contrasts.

Rubric and judge provenance. Both rubric families, both grader panels and both adjudication panels are language models. Reliability between panels is high (Table 1) and the adjudication panels were independent of the systems they judged and blind to how pairs were selected, but agreement is not correctness, and graders may share blind spots with the systems they grade.

The adjudication sample is band-limited. It covers pairs the clustering surfaced or nearly surfaced, so the recall figure of 0.907 is recall within reach of the lexical blocking rather than recall against all true duplicates in the library.

One reformatter, one prompt. The modality hardening in Table 8 is a property of that prompt as much as of the idea, and a reformatter constrained to preserve modality might produce atoms that cost less quality. We did not test one.

Composition destroys order. A skill's sequence carries meaning and the four-phase regrouping is a crude replacement. On procedural skills this should matter more than on checklist-shaped ones, and we do not separate the two.

The distractors are synthetic. The 459 skills added in Section 7 were written by a model, four atoms each, across twenty domains chosen to be unrelated to the evaluation tasks. A real 500-skill library would contain near-neighbours of the tasks that our distractors do not, so routing at true scale should be harder than Table 12 shows. The cost argument of Table 15 does not depend on this, since it turns on description count rather than description content.

Lexical retrieval throughout. No embedding model was available in the environment, so routing is BM25 with offline query expansion. Routing recall of 0.71 caps every result here, and the one failure we can characterise precisely is a vocabulary gap that no term weighting closes.

11 Conclusion

The composition idea is sound on a corpus built to contain the redundancy it removes and does not survive contact with a library written by people who were not thinking about composition. What survives is narrower and still useful: obligations are a good index key and a poor payload. Split each unit into the imperative and the teaching, compare only the imperative, keep every piece of teaching, and a heterogeneous skill library becomes de-duplicable at 0.817 precision where the previous method managed 0.067. Then print only the teaching, because the model already has the imperatives.

The runtime case rests on invariance. A composite's size is capped by construction and its quality did not move when we grew the library elevenfold, while whole-skill loading pays for every description in the library on every request and loads more skills the less sure it is. On a library of a few dozen skills that trade is not worth making. On a library of five hundred it is, and the reason has nothing to do with the composite getting better.

For anyone maintaining such a library, the obligation index is probably worth more at authoring time than at request time. It shows exactly where skills overlap, which is the question a maintainer has and cannot currently answer, and the number of overlaps it surfaces grows faster than the library does. Anyone promising quality parity with loading the skills outright should be asked for their rubric.

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A Reproducibility

Every artefact is deterministic given the seed. `tools/build_corpus.py` regenerates the synthetic corpus and its ground truth from the fragment library. `tools/stats_all.py` regenerates every contrast, effect size and reliability coefficient in this article, including the two-proportion test behind the merge-precision comparison of Table 9. The clustering of Table 2, the coverage sweep, the ablations of Section 4.5, the calibration of Section 5.3, the raw merge-adjudication counts of Table 9 and the fidelity audit of Table 8 are stored as computed results in `results/` rather than regenerated by a script in this release. `tools/verify_article.py` checks all 325 reported numbers, computed or stored, against those results and exits non-zero on any mismatch.

Selection protocol. The synthetic study's merging threshold was chosen on six skills and reported on the other six; its composition hyperparameters were chosen on four development tasks and its coverage and ablation figures reported on the remaining eight. The abstention floor was calibrated only on in-corpus and out-of-corpus probes, never on an evaluation task. The conjunctive merge rule of Section 5.3 was fitted on half the adjudication labels and reported on the other half. Offline expansions and atoms were generated by agents given only skill text, with instructions not to read anything else in the repository; they had no access to the task suites.